

# Conflict-Aware Cross-View Disaster Triage Across Spatial Observation Regimes [experiments]

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## ABSTRACT

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage. Aggregate F1 scores, however, hide the cases that matter most for deployment: examples where the single-view predictors disagree. This study introduces Conflict-Aware Evaluation (CAE), a protocol that reports overall F1, conflict-subset F1, and the conflict gain of cross-view fusion over the stronger single-view baseline. We apply CAE to two paired disaster settings: an Eaton/Altadena wildfire dataset with property-centered ground imagery and the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. The results show both the promise and the boundary of fusion. On the wildfire sensitivity split, cross-view fusion improves conflict F1 by 0.0388 over the best single-view baseline. On the leakage-controlled hurricane endpoint split, however, the gain becomes negative ( $\Delta_{\text{view}} = -0.0153$ ), even though aggregate F1 makes cross-view and street-only inference look nearly equivalent. A directional decomposition explains the switch: wildfire conflicts more often involve street-correct and overhead-wrong cases, while the broader hurricane split reverses this pattern. These findings support a simple evaluation shift: paired-view disaster models should be judged first on the conflict cases that expose their boundary conditions, not only on average-case scores.

## KEYWORDS

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

## 1 INTRODUCTION

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad spatial coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, but prior evaluation often reduces their value to one aggregate test metric. That view is too coarse for operational disaster triage. Many buildings are easy in both views. The harder and more consequential cases are those where the two views provide competing evidence.

This paper studies cross-view fusion as a conflict-resolution problem. We focus on paired examples that contain a ground image, an overhead patch, and a binary damage label. Instead of asking only whether paired-view fusion improves average F1, we ask whether fusion resolves the subset where the two single-view models disagree. That framing matters for geographic AI because observation conditions are not interchangeable. A close property inspection photograph and a panoramic street scene both count as ground imagery, but they encode different relationships between observer, target building, and surrounding landscape.

The study sits between several lines of disaster-vision and geospatial evaluation work. Satellite damage benchmarks such as xBD have made overhead-only building assessment a standard task [2], and later fusion models improve satellite-based damage assessment from multi-scale visual evidence [7]. Ground and street-view studies show that close-range imagery can capture damage cues that overhead imagery misses [8, 10, 11]. CVDIsaster is the closest paired-view benchmark for Hurricane Ian [4]. Cross-view geolocalization work has also established the broader term for matching ground and aerial views [6]; here the goal is not localization but damage triage under evidence conflict. CAE is related to selective prediction and disagreement-aware evaluation [1], but it does not claim a new risk-coverage estimator. Its contribution is a spatially grounded reporting unit for paired disaster views: the subset where the two observation regimes disagree, the single-view direction of correction, and the gain or loss from fusion. This complements spatial evaluation concerns about leakage and split design [5]. Recent generative work also studies how post-disaster ground views may be synthesized from satellite imagery when field observations are sparse [9].

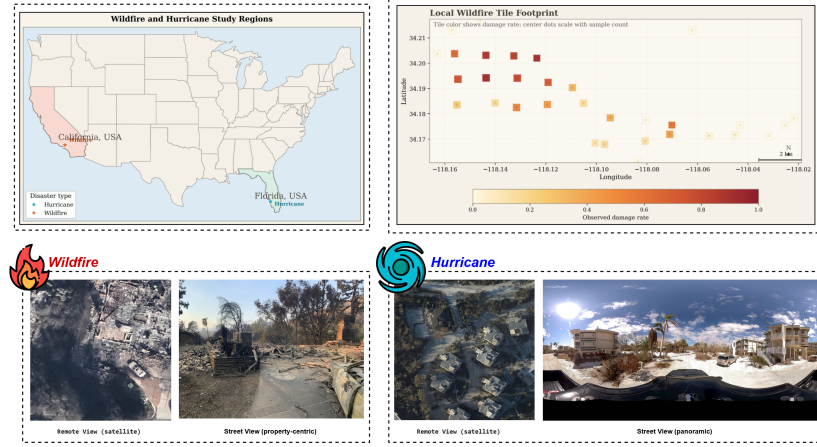
This study makes three contributions. First, it defines Conflict-Aware Evaluation (CAE), a three-metric protocol that reports full-test performance and conflict-subset behavior together. Second, it applies CAE to wildfire and hurricane paired-view triage and treats the hurricane grouped-clean split as the conservative endpoint control. This control produces the central negative result: fusion is not always the best conflict resolver after leakage is removed. Third, it introduces spatial observation regime as the explanation scale for this boundary. The analysis deliberately avoids a causal claim that disaster type alone determines which view should dominate. The evidence is better read as a regime-sensitive experiment: cross-view fusion helps most when single-view evidence conflicts, and the source of that help depends on how the ground view observes the target building.

## 2 DATA AND PROTOCOL

### 2.1 Paired-View Triage Settings

Let  $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$  be a paired disaster triage dataset, where  $x_i^{(s)}$  is a street or ground image,  $x_i^{(r)}$  is an overhead patch, and  $y_i \in \{0, 1\}$  is the binary damage label. We compare three modes: *street-only* (S), *remote-only* (R), and *cross-view* (CV). The cross-view model uses late fusion of ResNet18 features [3], including each embedding, their absolute difference, and their elementwise product. This simple fusion design keeps the evaluation focused on conflict behavior rather than architectural novelty.

(a) Spatial Observation Regimes and Geographic Context



**Figure 1: Study design and spatial observation regimes. The wildfire setting uses property-centered inspection imagery paired with overhead patches, whereas the hurricane setting uses panoramic ground imagery paired with overhead patches. This visual difference motivates the regime-aware interpretation tested by CAE.**

**Table 1: Evaluation settings and leakage controls.**

| Setting     | Positive class | Ground regime | Role            |
|-------------|----------------|---------------|-----------------|
| WF sens.    | affected+      | property      | main CAE        |
| H end.      | severe         | panoramic     | diagnostic      |
| H mod.+sev. | moderate+      | panoramic     | sensitivity     |
| H clean     | severe         | panoramic     | leakage control |

The wildfire setting uses an Eaton/Altadena paired dataset assembled for this project. Its ground imagery is property-centered inspection imagery paired with overhead patches. This observation regime often places the target building near the visual center of the ground image. The hurricane setting uses the Hurricane Ian CVD disaster benchmark [4], which pairs panoramic ground imagery with very high resolution overhead imagery. This observation regime often includes wider street context, adjacent buildings, and weaker target alignment. The comparison therefore mixes disaster type, geography, source, and collection protocol; those factors limit causal interpretation and motivate the conservative language used below.

Table 1 records the experimental settings and the control role of each split. This is important because the paper should not choose a split after observing which one makes the conflict story look strongest. The wildfire sensitivity split is used because it gives a larger and more informative disagreement subset. The hurricane grouped-clean endpoint split is included because it controls object-level leakage in the original endpoint partition.

## 2.2 Conflict-Aware Evaluation

For each example, let  $p_i^{(s)}$  and  $p_i^{(r)}$  denote the positive-class probabilities predicted by the street-only and remote-only models. CAE defines the conflict subset

$$C_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with  $\tau = 0.1$  in the main analysis. For each model  $m$ , CAE reports standard overall F1 and conflict F1 on  $C_\tau$ . It then computes the

cross-view conflict gain

$$\Delta_{\text{view}} = F1(\text{crossview}; C_\tau) - \max\{F1(\text{street}; C_\tau), F1(\text{remote}; C_\tau)\}. \quad (2)$$

This protocol exposes whether paired-view fusion resolves disagreement cases rather than only improving easy average cases.

We also decompose conflicts by correction direction.  $S \rightarrow O$  denotes cases where the street-only model is correct and the remote-only model is wrong;  $O \rightarrow S$  denotes the reverse. These fractions identify the stronger single-view arbitrator inside the conflict subset. Because a prior hurricane endpoint split had object-level leakage, we report the grouped-clean endpoint split as a required control rather than as an optional robustness run.

## 2.3 Implementation Details

All modes use the same triage implementation. Images are resized to 224 by 224 pixels and normalized for the selected backbone. The main runs use ImageNet pretrained ResNet18 encoders. Each view embedding is projected to 256 dimensions; the cross-view classifier receives the street embedding, overhead embedding, absolute difference, and elementwise product. Models are optimized with AdamW using learning rate  $10^{-4}$  for 10 epochs, and the checkpoint with the best validation F1 is evaluated on the held-out test split. This keeps the comparison between street-only, remote-only, and cross-view models tied to the same training protocol.

## 3 RESULTS

### 3.1 CAE Summary

Table 2 reports the core CAE results. If the analysis only reported overall F1, the conclusion would be incomplete: cross-view fusion is slightly better in wildfire, slightly better on two hurricane splits, and nearly tied with street-only on the grouped-clean hurricane endpoint split. CAE changes the question. It asks whether fusion helps on the cases where the two single-view models disagree.

**Table 2: Conflict-Aware Evaluation summary. Conflict F1 uses  $C_\tau$  with  $\tau = 0.1$ . WF is wildfire; H is hurricane. Gain is reported only for cross-view rows.**

| Setting     | M  | All    | C-F1   | Gain    | $n_c$ |
|-------------|----|--------|--------|---------|-------|
| WF sens.    | S  | 0.9392 | 0.6230 | –       | 435   |
| WF sens.    | R  | 0.9238 | 0.5237 | –       | 435   |
| WF sens.    | CV | 0.9473 | 0.6618 | +0.0388 | 435   |
| H end.      | S  | 0.9055 | 0.6842 | –       | 32    |
| H end.      | R  | 0.8969 | 0.5806 | –       | 32    |
| H end.      | CV | 0.9192 | 0.7647 | +0.0805 | 32    |
| H mod.+sev. | S  | 0.8738 | 0.7324 | –       | 128   |
| H mod.+sev. | R  | 0.8873 | 0.7742 | –       | 128   |
| H mod.+sev. | CV | 0.8915 | 0.7867 | +0.0125 | 128   |
| H clean     | S  | 0.9016 | 0.8101 | –       | 80    |
| H clean     | R  | 0.8385 | 0.6526 | –       | 80    |
| H clean     | CV | 0.9008 | 0.7949 | -0.0153 | 80    |

**Table 3: Directional conflict decomposition at  $\tau = 0.1$ .  $S \rightarrow O$  is street-correct and remote-wrong;  $O \rightarrow S$  is the reverse.**

| Split       | $n_c$ | $S \rightarrow O$ | $O \rightarrow S$ | Stronger view |
|-------------|-------|-------------------|-------------------|---------------|
| WF sens.    | 435   | 0.2713            | 0.1885            | street        |
| H end.      | 32    | 0.3750            | 0.3438            | near-tie      |
| H mod.+sev. | 128   | 0.1094            | 0.1328            | remote        |

The answer is useful because it is not uniformly positive. The wildfire sensitivity split has the clearest positive conflict gain: cross-view fusion reaches 0.6618 conflict F1, which is 0.0388 above the better single-view baseline. The original hurricane endpoint split also shows a large gain, but its small conflict subset ( $n = 32$ ) and leakage history make it diagnostic rather than primary evidence. The broader hurricane moderate-plus-severe split keeps a smaller positive gain. The grouped-clean endpoint split provides the most important boundary condition: cross-view fusion remains nearly tied with street-only overall (0.9008 versus 0.9016), but on conflict cases it falls below street-only by 0.0153. This negative result is not a failure of CAE; it is the reason CAE is needed. It shows that aggregate scores can hide the point at which fusion stops being the best conflict resolver.

### 3.2 Dominant-View Switch

The directional decomposition in Table 3 explains why the conflict gain changes across settings. On wildfire,  $S \rightarrow O$  conflicts are more common than  $O \rightarrow S$  conflicts (0.2713 versus 0.1885), so the ground image is the stronger single-view arbitrator. This matches the observation regime: property-centered ground imagery often places the target building near the image center and exposes facade, debris, and local damage cues that an overhead patch can miss.

The hurricane settings behave differently. On the endpoint split, the two directions are near tied (0.3750 versus 0.3438), which helps explain the large but unstable positive gain. On the broader hurricane split, the ordering reverses (0.1094 versus 0.1328), making the overhead view the stronger single-view arbitrator. This does not mean that hurricanes favor overhead imagery in general. It means that panoramic ground imagery, with wider scene context and weaker target alignment, can behave differently from property-centered inspection imagery. The stable finding is the switch itself: dominant-view behavior changes with spatial observation regime and split definition.

### 3.3 Sensitivity Checks

The conflict-centered interpretation is not only a threshold artifact. A separate sweep at  $\tau = \{0.1, 0.3, 0.5\}$  keeps CV strongest on wildfire conflict accuracy (0.8349, 0.7821, 0.7538) and on the hurricane moderate-plus-severe conflict subset (0.7097, 0.6400, 0.6190). Under the reporting threshold  $\tau = 0.1$ , CV accuracy on conflict cases is 0.7885 for the wildfire sensitivity split, compared with 0.7356 for S and 0.6529 for R. On the hurricane moderate-plus-severe split, the corresponding values are 0.7500, 0.7031, and 0.7266. Label-independence permutation tests are below  $10^{-4}$  in both settings.

Multi-seed validation supports the base ordering without changing the main claim. Wildfire CV reaches  $0.9689 \pm 0.0011$  validation F1, while S and R reach  $0.9657 \pm 0.0001$  and  $0.9654 \pm 0.0015$ . Hurricane CV reaches  $0.9455 \pm 0.0063$ , compared with  $0.9433 \pm 0.0012$  and  $0.8959 \pm 0.0065$ . Backbone checks add a practical caution: ResNet18 and ResNet50 remain strong, while generic CLIP and DINOv2 features underperform the supervised CNN baselines in this disaster-specific triage setting. These checks support the evaluation lens, but they are not the primary contribution. The primary contribution is the positive-and-negative conflict pattern exposed by Tables 2 and 3.

## 4 DISCUSSION

The experiment supports CAE as a useful evaluation protocol for paired-view disaster triage. Aggregate F1 alone would make the models look similar in some settings. CAE reveals how they behave on disagreement cases and shows that cross-view value can shrink, grow, or become negative relative to the stronger single-view model after leakage control.

### 4.1 Boundary Conditions Are Results

The grouped-clean hurricane result is the most important stress test in the paper. It shows that more visual evidence does not automatically produce a better decision once the evaluation split removes object-level leakage. This is not a weakness of the study. It is the kind of result that deployment-oriented GeoAI needs: a condition under which a plausible multimodal assumption stops holding. In disaster response, the costly mistake is not only using too little information. It is also trusting a fused model in the subset where a simpler single-view model is already the better arbitrator.

This framing also clarifies the role of the original hurricane endpoint split. The endpoint result remains useful for benchmark continuity, but it should not carry the main conclusion once the grouped-clean split is available. The clean split sets the conservative boundary; the endpoint split shows why that boundary is needed. Reporting both prevents the paper from choosing the partition that makes fusion look strongest.

The same principle applies to the wildfire dataset. Its positive result is credible only if the dataset is not treated as a black box. The paper artifact will therefore release wildfire split manifests and a data-card summary with train/validation/test counts, class balance under the sensitive label mapping, objectid grouping, image-source fields, coordinate fields, missing-pair filtering rules, and access notes for the raw imagery. The raw images are not tracked

directly in the GitHub repository, so this metadata release is the reproducible unit for verifying the split and reconstructing the paired-view benchmark.

## 4.2 Spatial Observation Regime

The dominant-view switch suggests that *ground imagery* is too broad a category for paired-view disaster assessment. Property-centered ground imagery and panoramic street imagery observe target buildings differently. The former often aligns the camera with the damaged structure and can expose close facade and debris cues. The latter provides wider street context, adjacent buildings, and scene-level evidence, but it may weaken the correspondence between the ground view and the labeled overhead footprint.

Spatial observation regime names this difference. It is not a new sensor type or model architecture. It is the relationship among observer, target building, viewpoint geometry, and surrounding context. CAE makes that relationship measurable because it asks which single view corrects the other when they conflict. In the wildfire setting, the property-centered view more often corrects the overhead view. In the broader hurricane setting, the overhead view more often corrects the panoramic view. The mechanism is therefore not simply “street versus satellite”; it is how each view observes the target under a specific collection regime.

This concept should transfer beyond the two datasets used here. Any multimodal GeoAI task that pairs overhead imagery with ground, mobile, or generated views faces the same problem: the modalities may agree on easy cases and disagree where target alignment, occlusion, context, or collection timing matters. CAE turns those disagreements into a reportable unit rather than averaging them away.

This wording also avoids a common shortcut in multimodal evaluation. A dataset description may say that it contains satellite and street-view imagery, but the operational question is more precise: does the ground view observe the labeled object, the surrounding block, or only a nearby scene? A benchmark that records this observation relation can explain why a fused model succeeds in one place and fails in another. Without that record, a single aggregate score can make two very different sensing situations look comparable.

## 4.3 Implications and Next Steps

CAE suggests a practical triage policy. If street-only and remote-only predictions agree with high confidence, a response system can often use the cheaper agreement decision. If they disagree, the case becomes a candidate for cross-view fusion or human review. The exact threshold should be tuned for each deployment, but the principle is stable: cross-view reasoning is most valuable when the evidence is already in tension.

The same view can guide dataset design. A future benchmark should collect property-centered and panoramic ground imagery for the same disaster and the same target buildings. That design would remove many confounders in the present comparison and would let CAE test whether target alignment, rather than hazard type or data source, drives the dominant-view switch. It would also support spatial diagnostics, such as mapping where conflict density clusters across neighborhoods, but those analyses should

be treated as follow-up work rather than the main contribution of this study.

The main claim is narrower: a paired-view benchmark should first report whether fusion resolves existing view disagreements.

## 5 CONCLUSION

This work reframes paired-view disaster triage as conflict-aware evaluation. Across wildfire and hurricane paired-view settings, average F1 hides the cases that determine whether fusion is useful in practice. CAE exposes those cases. It shows a positive wildfire conflict gain, a negative grouped-clean hurricane gain, and a dominant-view switch that follows the spatial observation regime. The resulting lesson is practical: disaster benchmarks should report conflict-subset behavior alongside aggregate scores. Doing so makes multimodal fusion more falsifiable, more spatially interpretable, and more useful for deployment decisions.

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